



PHARMACOLOGY 1

FACULTY OF NURSING

- **Pharmacodynamics** (Gr. **Pharmacon**: drug; **dynamis**: power):
 - it covers all the aspects relating to **‘what the drug does to the body’**.
- **It is the study of drugs—**
 - ✓ their **mechanism** of action,
 - ✓ pharmacological actions,
 - ✓ adverse effects.

Types of Drug Action

1. Stimulation

2. Depression

3. Irritation

4. Replacement

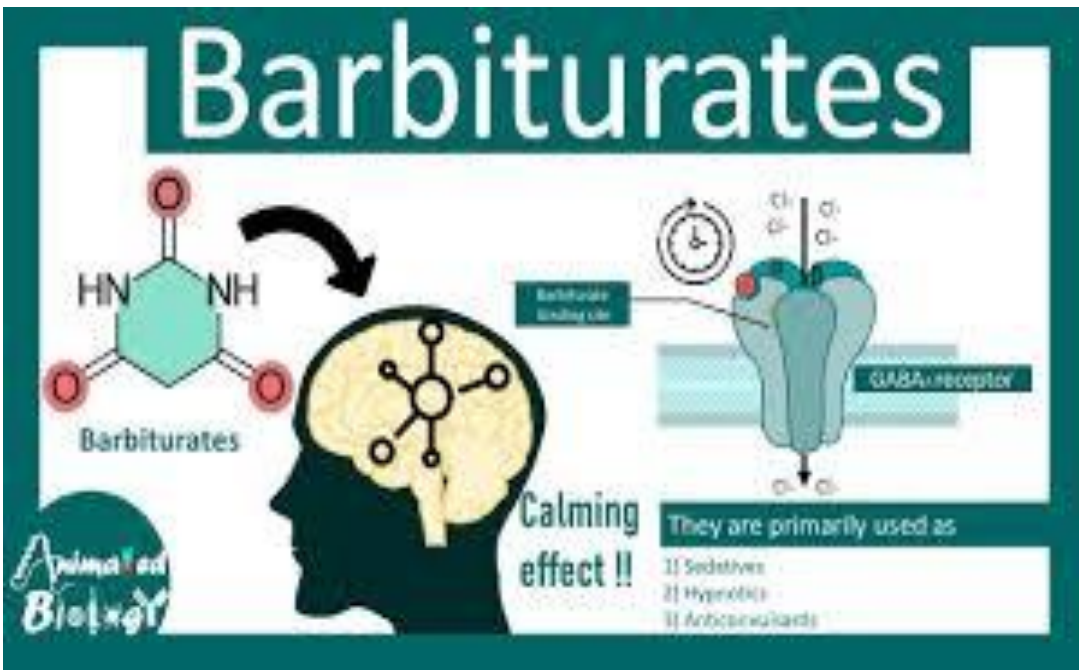
5. Cytotoxic

1. Stimulation:



Some drugs act by **increasing the activity** of specialized cells, **e.g., adrenaline stimulates** the heart increasing heart rate and **force of contraction**.

2. Depression:

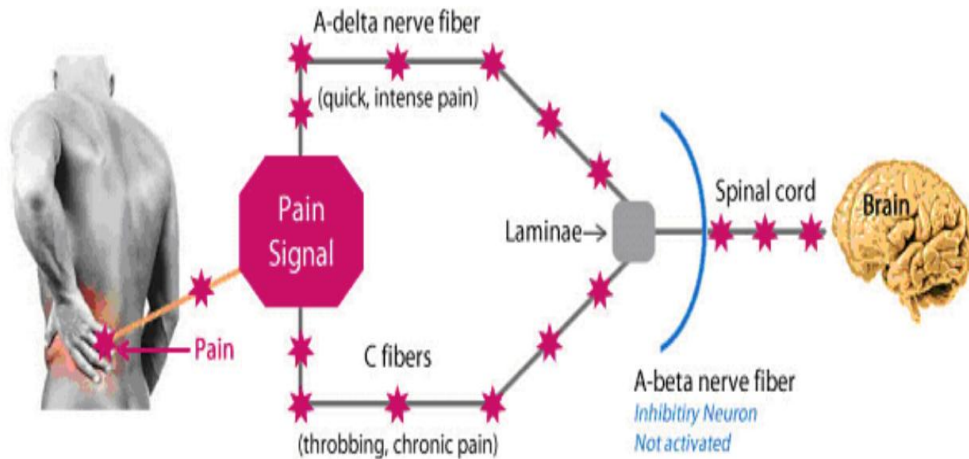


Some drugs act by **decreasing** the activity of specialized cells,

e.g., alcohol, **barbiturates**, general anesthetics, etc. **depress** the central nervous system.

3. Irritation:

Pain is Perceived



Certain agents on **topical application** can **irritate** the skin and adjacent tissues.

When an agent on application to the skin relieves deep-seated pain, it is known as a **counterirritant (e.g., eucalyptus oil, methyl salicylate)**.

They are useful in sprains, joint pain, myalgia, etc.

4. Replacement



When there is **a deficiency of endogenous** substances, they can be replaced by drugs,

e.g., insulin in diabetes mellitus.

5. Cytotoxic



Drugs are **selectively toxic** to the infecting **organism/cancer** cells, **e.g., antibiotics/ anticancer** drugs.



Mechanism of Drug Action



**Nonreceptor-
mediated
Mechanisms**



**Receptor-
mediated
Mechanisms**

Nonreceptor- mediated Mechanisms

1. By physical action

2. By chemical action

3. Through enzymes

4. Through ion channels

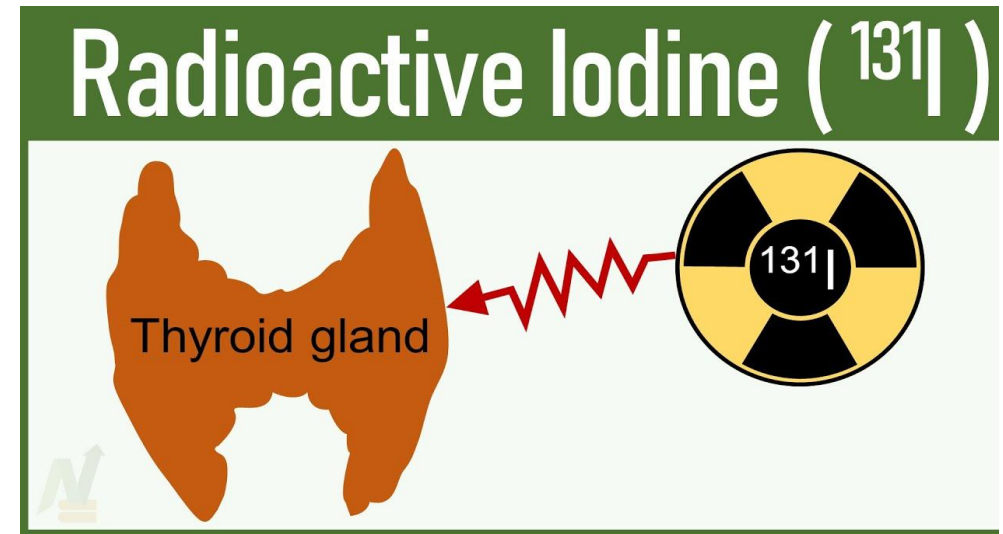
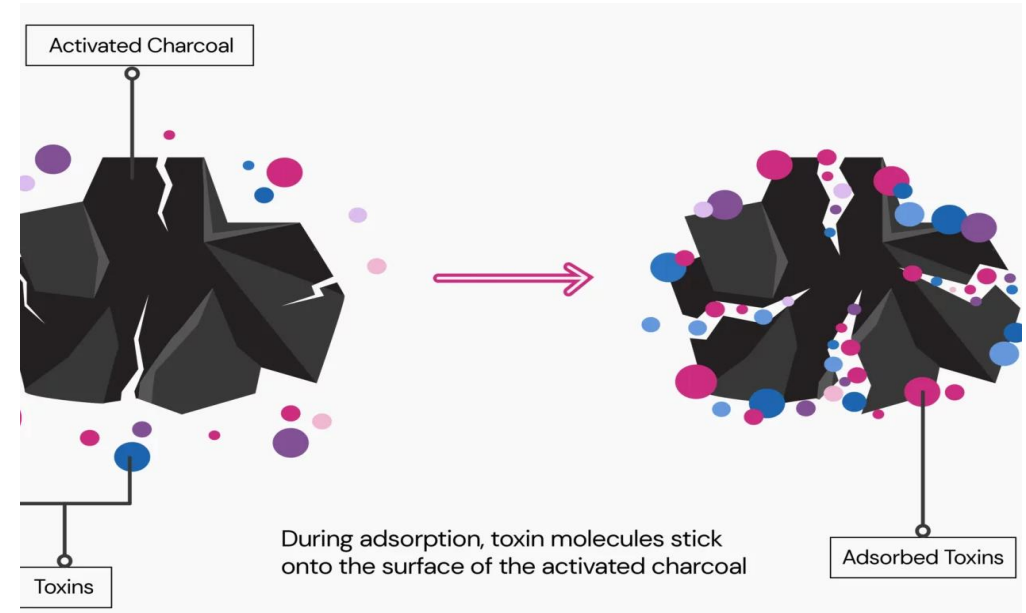
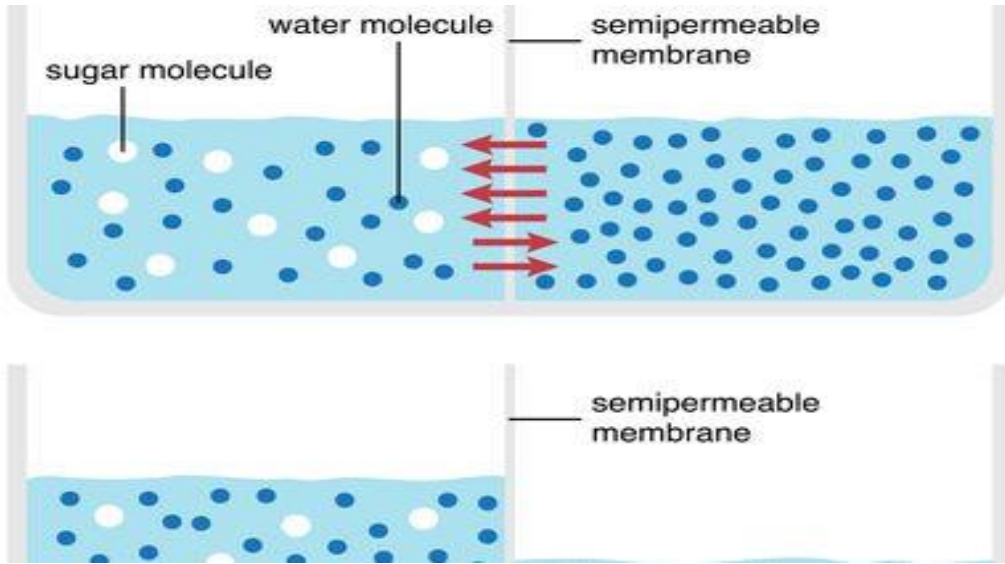
5. Through antibody production

6. Transporters

7. Others

1. By physical action:

- a. Osmosis:** Some drugs act by exerting an osmotic effect, **e.g.**, 20% **mannitol** in cerebral edema.
- b. Adsorption:** **Activated charcoal** adsorbs toxins; hence it is used in the treatment of drug **poisoning**.
- c. Demulcent:** Cough syrup produces **a soothing effect** in pharyngitis by coating the inflamed mucosa.
- d. Radioactivity:** **Radioactive isotopes** emit rays and destroy the tissues, **e.g.**, ¹³¹I in **hyperthyroidism**.



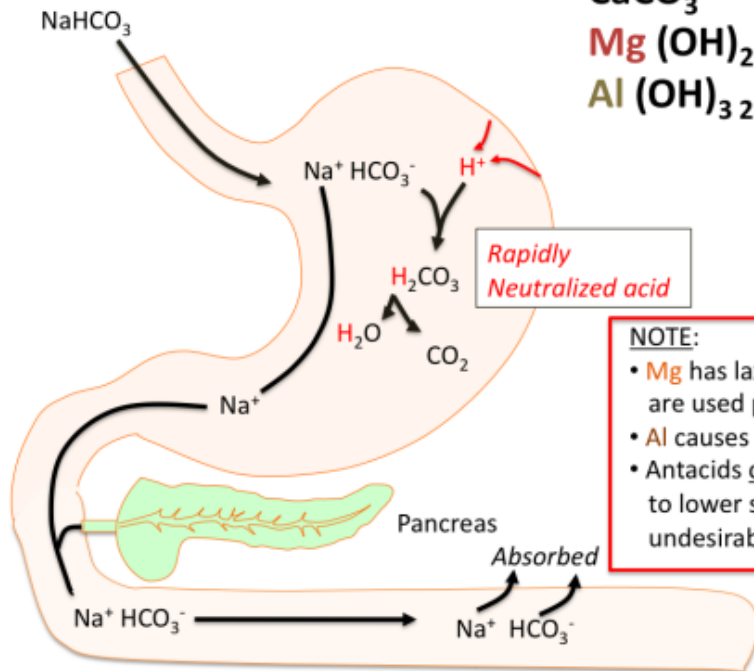
2. By chemical action:

a. **Antacids** are **weak bases**; hence they **neutralize acid** in the stomach in peptic ulcer.

b. **Metals** like **iron**, **copper**, and **mercury**, are eliminated from the body with the help of **chelating agents**.

They **trap** the metals in their **ring structure** and form **water-soluble complexes**, which are rapidly **excreted** from the body.

Na Bicarbonate



Other Antacids:

CaCO_3 (H^+ binds w/ CO_3^{2-})

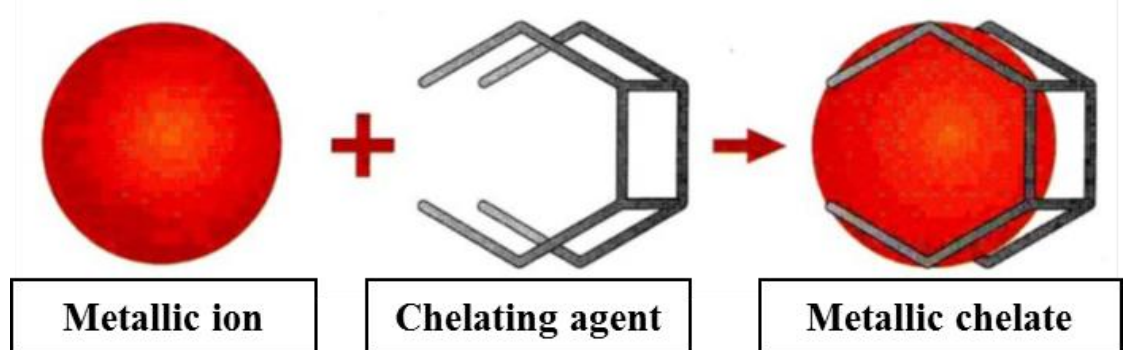
$\text{Mg}(\text{OH})_2$ (H^+ binds w/ OH^-)

$\text{Al}(\text{OH})_3$ (H^+ binds w/ OH^-)

NOTE:

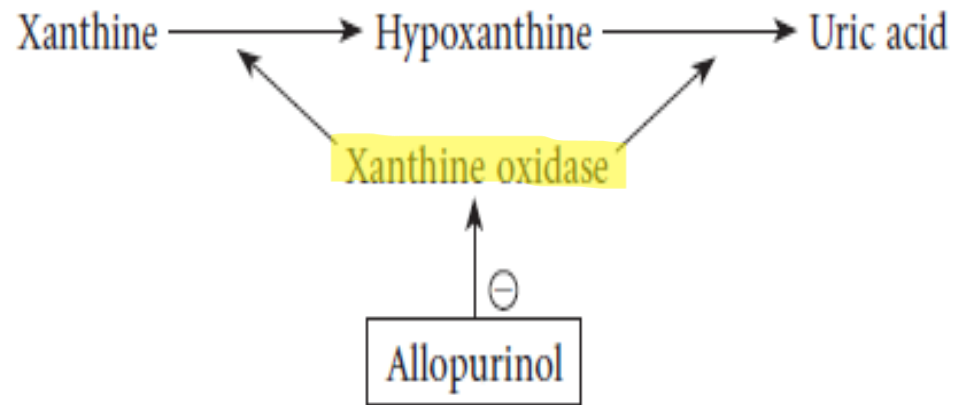
- **Mg** has laxative effects (laxatives w/ Mg are used prior to endoscopy)
- **Al** causes constipation
- Antacids combining Al & Mg are used to lower stomach acid w/o producing undesirable constipation or diarrhea

Adapted from Lullmann H et al. (2005)



3. Through enzymes:

- Almost all biological reactions are carried out under the catalytic influence of enzymes; hence, **enzymes** are a very important **target** of drug action.
- Drugs can either **increase** or **decrease** the rate of enzymatic media

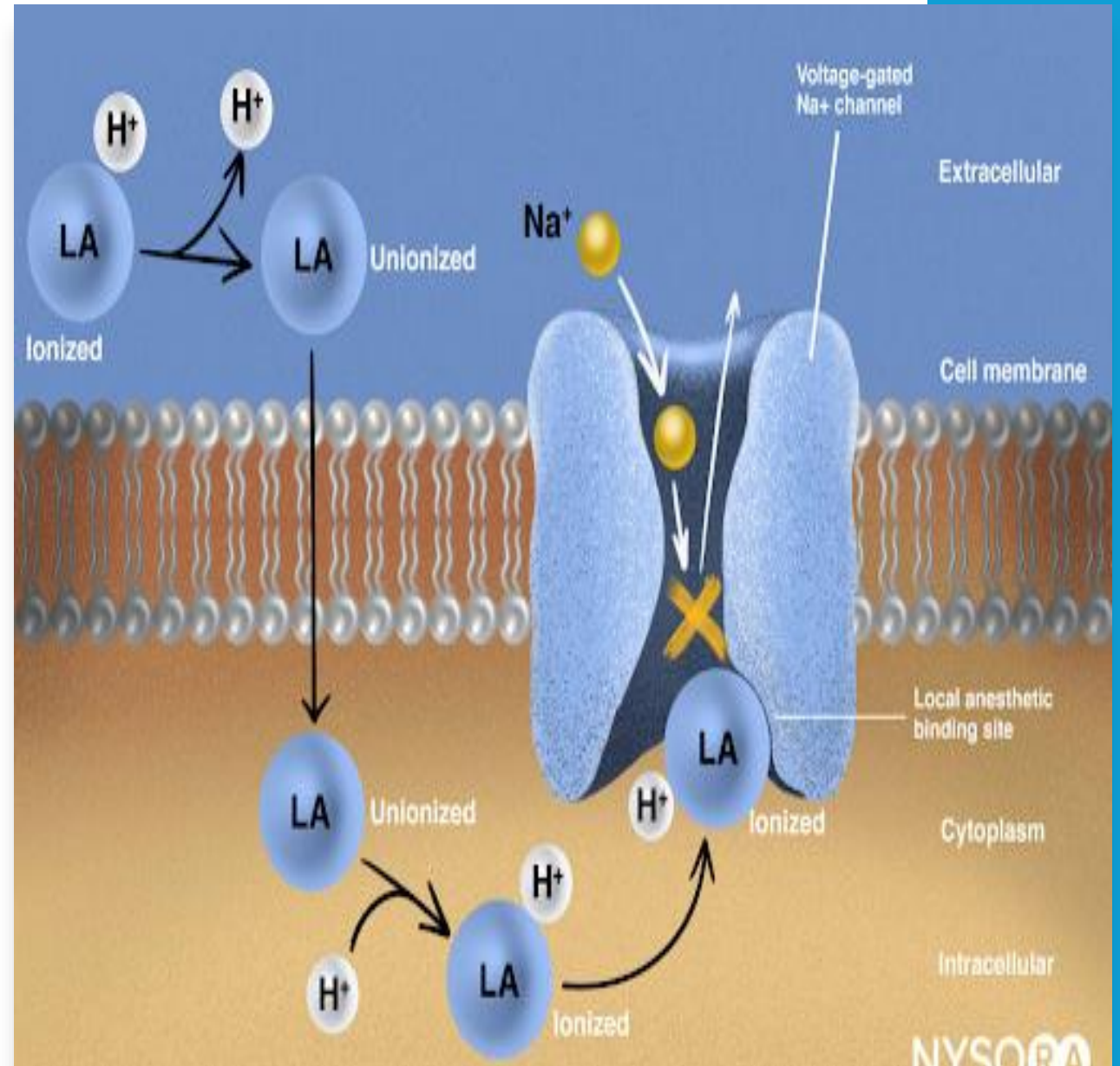


- **Drug action via enzyme inhibition:**

- **Xanthine** and **hypoxanthine** are **oxidized** to uric acid by the **enzyme xanthine oxidase**, which is inhibited by **allopurinol**.
- Allopurinol is used in the treatment of chronic **gout** to reduce the synthesis of uric acid.

4. Through ion channels

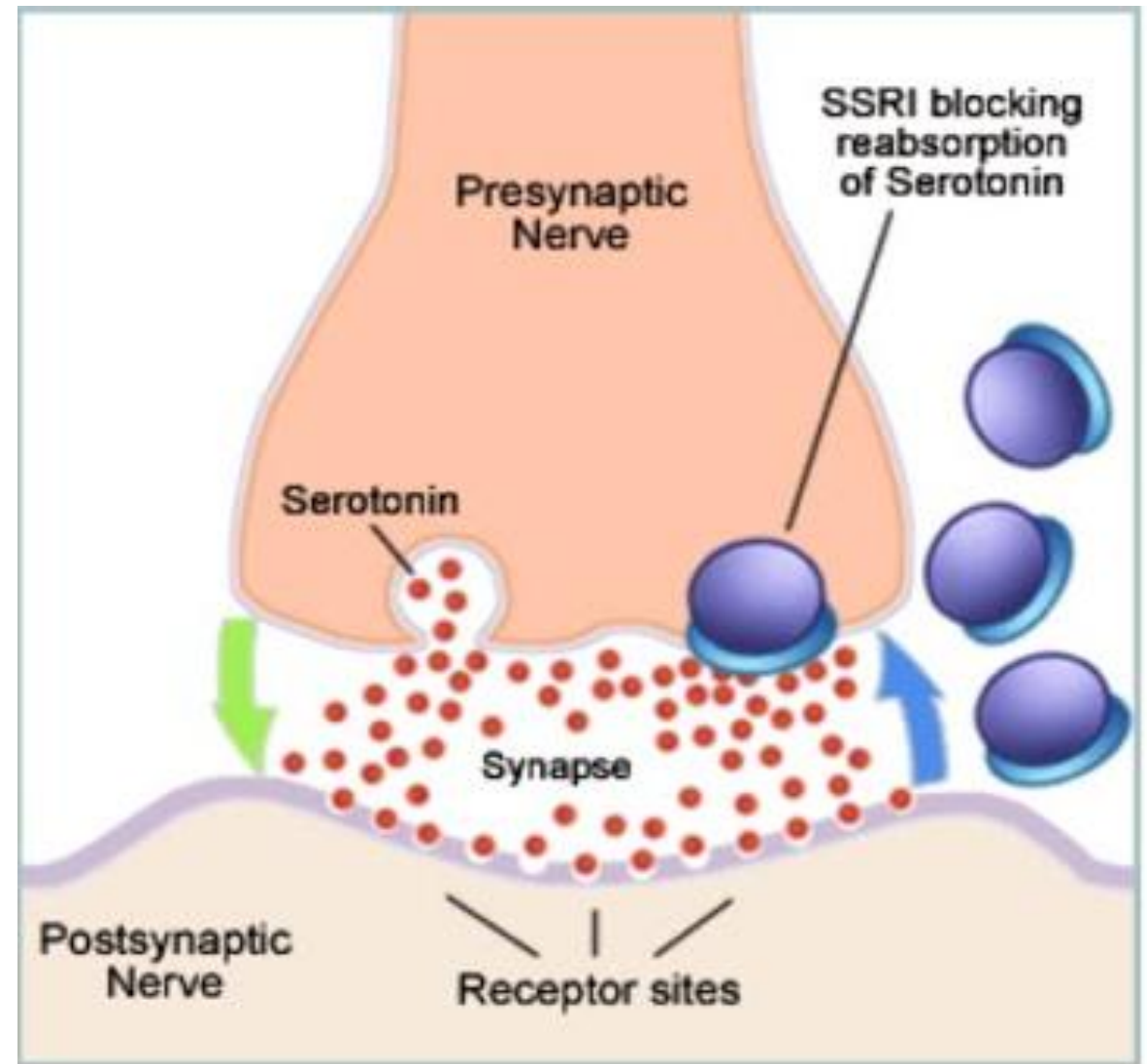
- Some drugs directly **bind to ion channels** and alter the flow of ions, **e.g.**
- **local anesthetics block sodium channels** in the neuronal membrane to produce **local anesthesia**.



6. Transporters

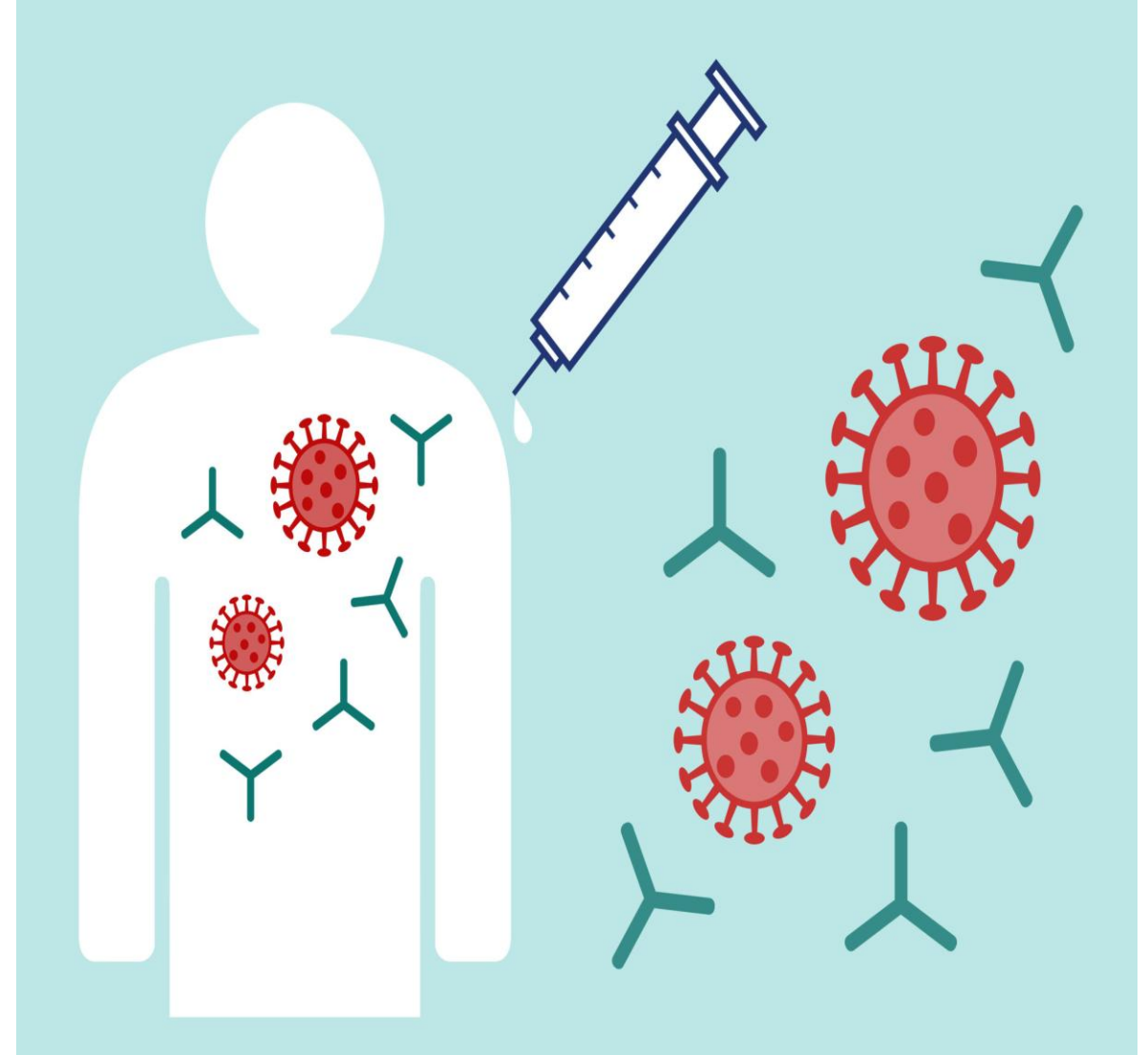
Some drugs produce their effect by binding to **transporters**.

Selective serotonin reuptake inhibitors (SSRIs) bind to **5-HT transporter** block **5-HT** reuptake into neurons antidepressant effect.



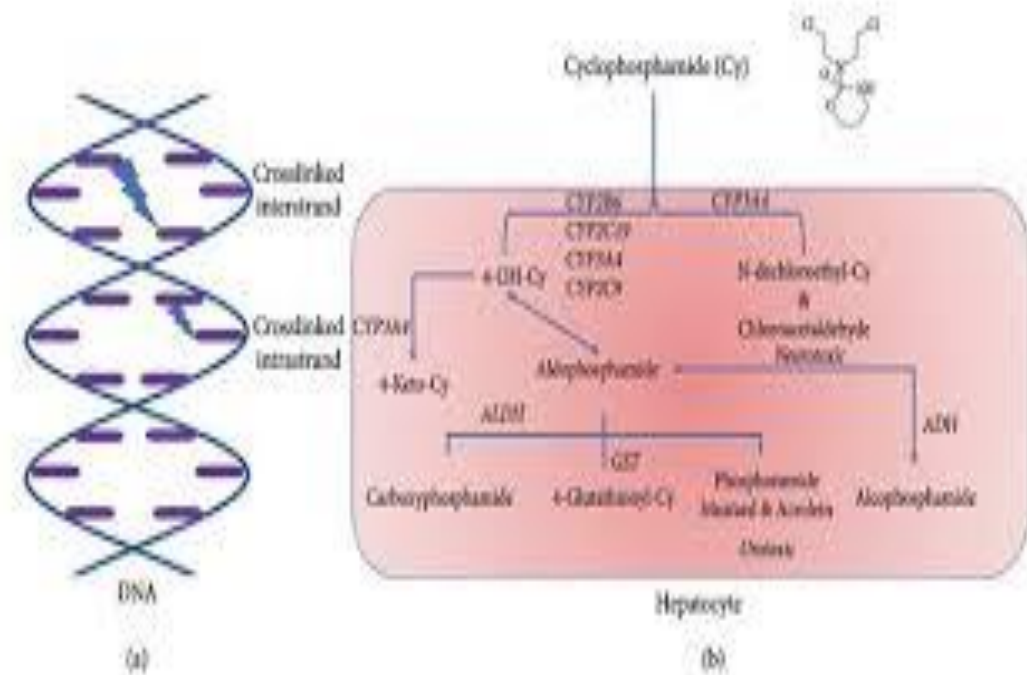
5. Through antibody production:

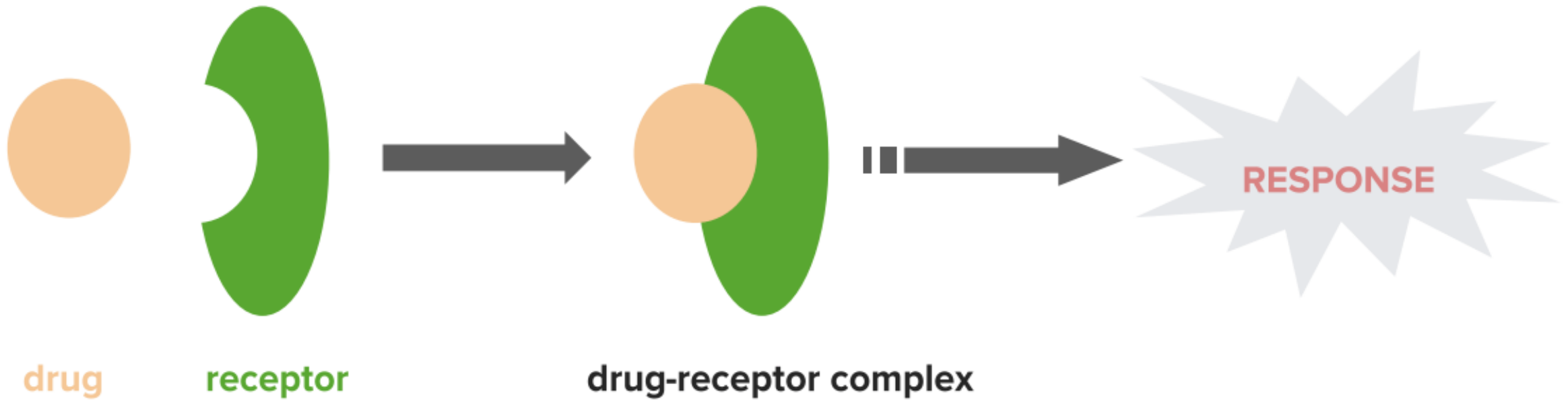
Vaccines produce their effect by stimulating the formation of **antibodies**, **e.g.**, vaccines against tuberculosis (BCG)



7. Others:

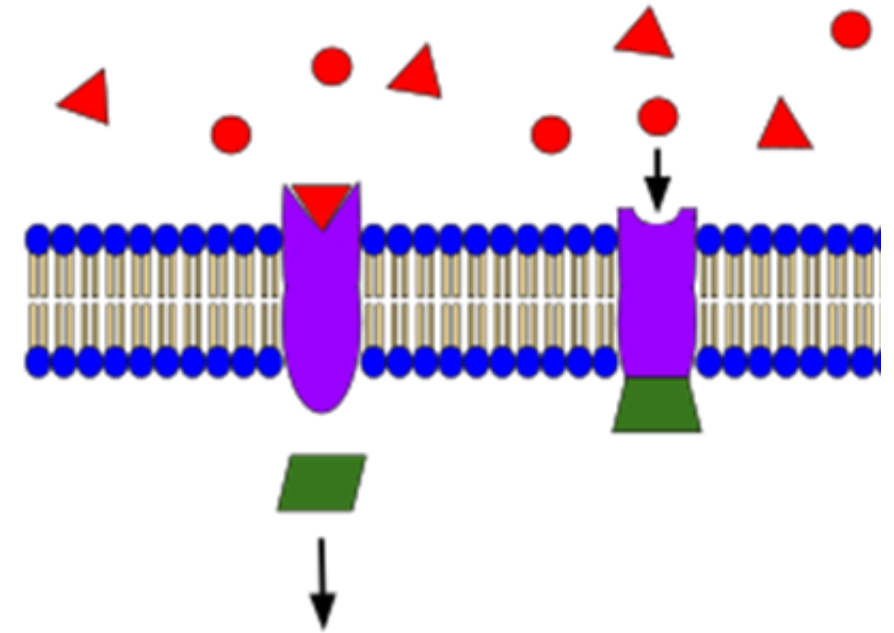
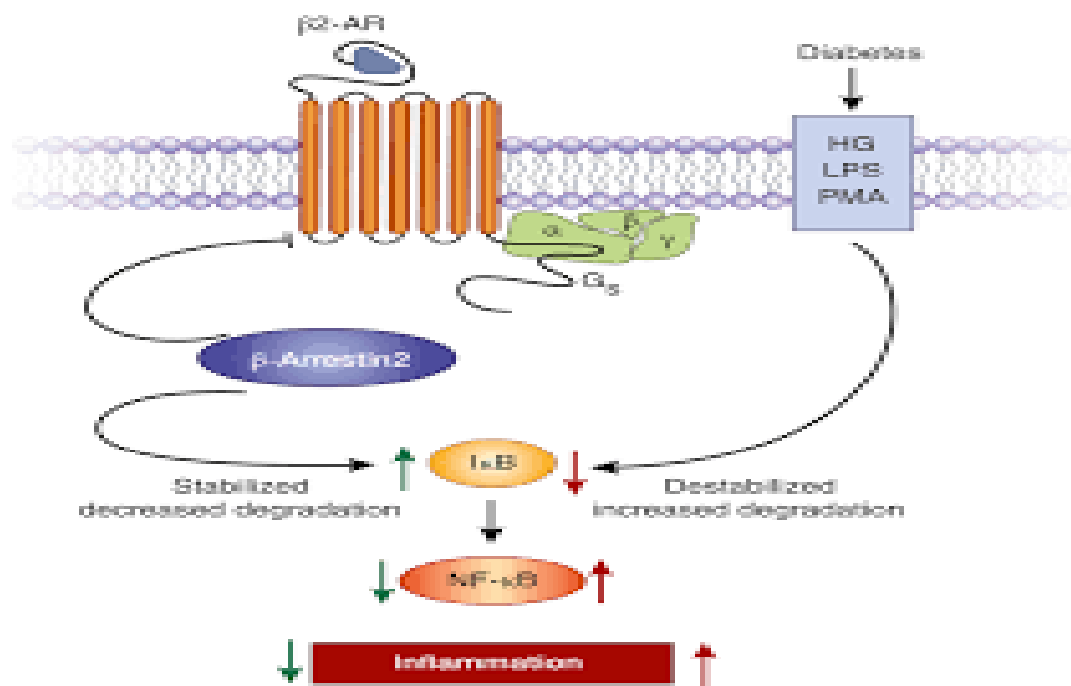
Anticancer drugs like **cyclophosphamide** produce their effect by binding to **nucleic acids**





Receptor-mediated Mechanisms

- Receptors are **macromolecules** present either on the **cell surface**, **cytoplasm**, or in the nucleus with which the drug binds and interacts to produce cellular changes.



- **For example**, adrenergic receptors, cholinergic receptors, opioid receptors, etc.

LOCK & KEY MODEL

